

Effect of phase-angle jump in Impedance-based Fault Location Methods for Transmission and Distribution Systems

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Abstract—The effect of phase-angle jump on impedance-based fault location methods for transmission and distribution systems is studied in this paper. The influence of fault resistance in simple impedance, simple reactance, Takagi and modified Takagi fault location methods is evaluated in four cases, and a correlation between fault resistance and phase-angle jump is stated. Three IEEE 4-Node test feeders are used to carry out faults with different fault resistance in high-voltage and low-voltage transmission and distribution lines, the phase-angle jump is computed by comparing the pre-fault and during fault voltage phasor in the faulted phase, and the fault location error is computed according to IEEE C37.114 standard. The difference in the phase-angle jump in high-voltage and low-voltage lines is also shown. From the simulation results, it is shown that the phase-angle jump is a useful feature to detect and locate faults. The inclusion of the phase-angle jump in fault location methods can be an effective way to improve the accuracy of fault location algorithms.

Index Terms—fault location, phase shift, power quality monitoring, unbalanced faults, voltage sags.

I. INTRODUCTION

Fault are inevitable phenomena that occurs very frequent in transmission and distribution systems. Faults are caused by a variety of reasons and cause a great impact on industrial and commercial users [1]. In the same way, fault location is a critical issue in transmission and distribution systems. An accurate fault location allows the system operator to isolate and repair the abnormal situation that caused the fault, reducing the impact on the user, guaranteeing proper system operation, and improving system reliability. Voltage and current waveforms are usually used to locate faults. Also, system information such as line impedance, equivalent source impedance, and system type is used to have better performance in locating faults. There are a variety of fault location algorithms that give results with acceptable accuracy, but system characteristics put a challenge to those algorithms. Line conditions like non-homogeneity, inaccurate system input data, current transformer saturation, fault resistance, untransposed lines, and so on are challenging for location purposes. For that reason, improvements on fault location algorithms that take into consideration common conditions like non-homogeneous systems and fault resistance are needed.

A feature that takes into consideration non-homogeneity of the system and fault resistance is the phase-angle jump (or phase shift). The phase-angle jump is a voltage sag characteristic that is suggested to be included in voltage sag analysis [2]. The work done by [3], [4] is mainly focused on the effects of PAJ on sensitive equipment, including procedures for testing the tolerance of equipment to this voltage sag characteristic. The work done by [5] mentioned the influence of fault resistance in the phase-angle jump, but do not advance in characterize the impact of this feature in fault location techniques. It was stated in [6] that algorithms that take into consideration the phase-angle jump in their analysis yield more accurate results. For that reason, a characterization of the effect of the phase-angle jump in fault location algorithms is needed. In this paper, the impact of the phase-angle jump in fault location of some traditional fault location algorithms is studied. Impedance-based fault location algorithms that do not use the equivalent source impedance information of the system are analyzed since it is no reasonable to assume that the equivalent source impedance is known. The equivalent source impedance change with the system operation along the day and system reconfiguration, which is usual in the operation of distribution systems [7]. Also, the phase-angle jump must be considered in distribution lines since the majority of distribution lines are non-homogeneous.

By characterizing the effect of the phase-angle jump in the fault location and its relation with fault impedance and non-homogeneous systems, measures can be taken to reduce the impact of this inevitable effects in fault location practice. This impact is studied in a distribution system changing the fault location and fault impedance to efficiently show that these two characteristics are closely related to the phase-angle jump. Faults in different voltage levels are also analyzed to see some other application opportunities of the phase-angle jump.

This paper is organized as follows: the phase-angle jump influence in fault location and fault resistance is presented in Section II. In section III the phase-angle jump characterization with different fault resistance is validated in a simulation test system. Finally, conclusions and future work on the proposed results are stated.

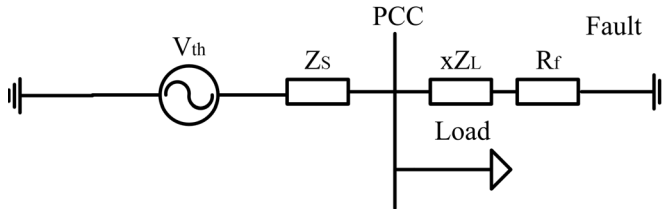


Fig. 1. Voltage divider model.

II. FAULT LOCATION AND PHASE-ANGLE JUMP INFLUENCE

In the next subsections, the phase-angle jump concept is explained, and his influence on fault location error and fault location in non-homogeneous systems is analyzed.

A. Phase-angle jump

A voltage sag characteristic introduced in [2] is the phase-angle jump or phase shift. A phase shift is defined by the authoritative dictionary as "(4) The displacement in time of one waveform relative to another of the same frequency and harmonic content" [8]. This phenomenon is studied because it affects some sensitive equipment and it is present in the majority of voltage sags [3], [4]. The phase shift is caused by the difference in the X/R ratio between the equivalent source and the line in a transmission or distribution system. The phase-angle jump not only affects sensitive equipment but also has a crucial role in faults with fault impedance different of zero. For balanced faults with fault impedance, the voltage divider model for radial systems of Fig. 1 is the following [5]:

$$V_{sag} = \frac{Z_L + R_f}{Z_S + Z_L + R_f} \quad (1)$$

where V_{sag} is in per-unit, Z_L is the line impedance up to the fault, Z_S is the impedance of the equivalent source and R_f is the fault resistance. From (1) the phase-angle jump is the angle of V_{sag} and can be computed as follows:

$$paj = \arctan \frac{X_L}{R_L + R_f} - \arctan \frac{X_S + X_L}{R_S + R_L + R_f} \quad (2)$$

Where paj is the phase-angle jump, X_L and X_S are the reactive part of the line impedance and equivalent source impedance respectively, and R_L and R_S are the resistive part of the line impedance and equivalent source impedance respectively. From (2) it can be observed that the fault resistance significantly affect the phase-angle jump value, so it can be used to detect high impedance faults. In the case of single phase to ground faults, the voltage divider model is [5]:

$$V_{sag} = \frac{2Z_{L1} + Z_{L0} + 2Z_{S1} + Z_{S0} + 3R_f - 3Z_{S1}}{2Z_{L1} + Z_{L0} + 2Z_{S1} + Z_{S0} + 3R_f} \quad (3)$$

Where Z_{L1} is the positive-sequence line impedance, Z_{L0} is the zero-sequence line impedance, Z_{S1} is the positive-sequence equivalent source impedance and Z_{S0} is the zero-sequence equivalent source impedance. From (3), it is evident that the phase-angle jump equation gets a little bit more complicated, but it is the angle of V_{sag} in (3), which involve the zero-sequence impedances of the equivalent source and the line.

B. Impedance-based fault location algorithms

Fault location is critical in service restoration and has a crucial impact on transmission and distribution system reliability. Since distribution systems continually change due to operational situations [7], fault location algorithms that do not require equivalent source impedance information are important. Also, the result of fault location algorithm is susceptible to the accuracy of the equivalent source impedance value. Thus, fault location algorithms that do not require the equivalent source impedance value are preferred. The impedance-based fault location algorithms analyzed in this paper are: simple impedance, simple reactance, Takagi method and modified Takagi method. The methods above require inputs: line impedance, line length, voltages, and currents. The equations for those impedance-based methods can be found in [7], [9]. One of the important assumptions that all the mentioned impedance-based fault location algorithms make is that the system is homogeneous. This assumption does not hold when the fault impedance is different from zero. In the following section, a case study is to be shown in which the fault impedance effect is shown.

C. Fault detection in lateral feeders and non-homogeneous distribution systems

Besides its importance in voltage sag analysis, phase-angle jump computation can be useful in detecting faults in low voltage lines and lines that have underground sections. Using the phase-angle jump concept a first fault detection can be made. It is known that underground cables have an important component of capacitive reactance, so the X/R ratio is significantly different compared with an overhead line. Thus, through a proper characterization of each section of a non-homogeneous distribution system, faults can be properly detected using the phase-angle jump measured at the point of common coupling. Aside from its residual voltage, high impedance faults can be differentiated from faults in the low voltage line of a transformer by computing the phase-angle jump during the event. As will be illustrated in the next section, faults with impedance different from zero are characterized by high phase-angle jumps, and the contrary happens with faults in low voltage lines. Also, faults in lateral feeders can also be identified if the conductor size of the feeders is different (so, the X/R ratio of feeders is different).

III. CASE STUDY

Three distribution systems are used to prove the effectiveness of the phase-angle jump in fault location. The systems used were taken from IEEE4-Bus test feeders. In the first case the sensitivity to fault impedance of simple impedance, simple reactance, Takagi and modified Takagi methods is checked. Next, other two test systems are used to show the usefulness of using the phase-angle jump values for fault location.

A. Fault location accuracy with fault impedance different from zero

The first 4-Node test feeder is used to evaluate the sensitivity of fault location methods to phase-angle jumps. The system

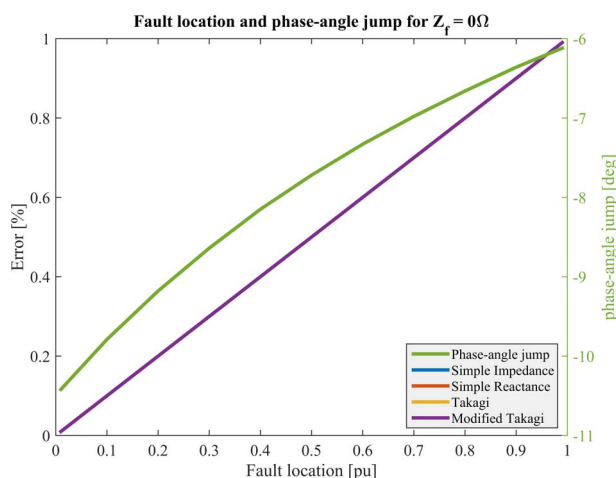


Fig. 2. Fault location and phase-angle jump for fault impedance of 0Ω .

consists of an equivalent source with Thevenin equivalent $Z_{th} = 0.0598 + j0.7974\Omega$, an overhead distribution line of 2000 ft, a 6 MVA step-down 12.47/4.16kV transformer, an overhead low voltage line of 2500 ft and a balanced load of 6 MVA, 0.9 lagging power factor. The distribution line configuration and system parameters can be found in [10]. Faults along high voltage line of the system were simulated with fault impedances equal to 0Ω , 0.5Ω , 1Ω , and 3Ω were simulated to evaluate the accuracy of fault location algorithms. Also, the phase-angle jump was computed to see its influence in fault location accuracy. The phase-angle jump was calculated by comparing the angle of the phasor during the fault with the pre-fault phasor.

1) *Fault location accuracy with $Z_f = 0\Omega$* : in this case, the simulated faults have no fault impedance and the fault location was computed according to [7]. The results are shown in Fig. 2. The fault location error, expressed in percentage with respect to the total line length, was computed using equation (2) of [7].

From Fig. 2, the left axis corresponds to the error of fault location algorithms and the right axis correspond to the computed phase-angle jump. It can be observed from Fig. 2 that the location error of all algorithms is less than 1% in all cases. Also, it is observed that the phase-angle jump is relatively small, taking into consideration that phase-angle jump in distribution systems can be up to 60° [5], so the error introduced by the phase-angle jump is small.

2) *Fault location accuracy with $Z_f = 0.5\Omega$* : The results of this case are shown in Fig. 3. From Fig. 3, the left axis corresponds to the error of fault location algorithms and the right axis correspond to the computed phase-angle jump. It can be observed that the error is significant for simple impedance (blue line) and simple reactance (red line) methods are significant, more than 10% in all cases. Takagi and modified Takagi methods are less affected by the introduction of the fault impedance, but the error is more than 1% in all cases. For this case, the phase-angle jump is significant (more than

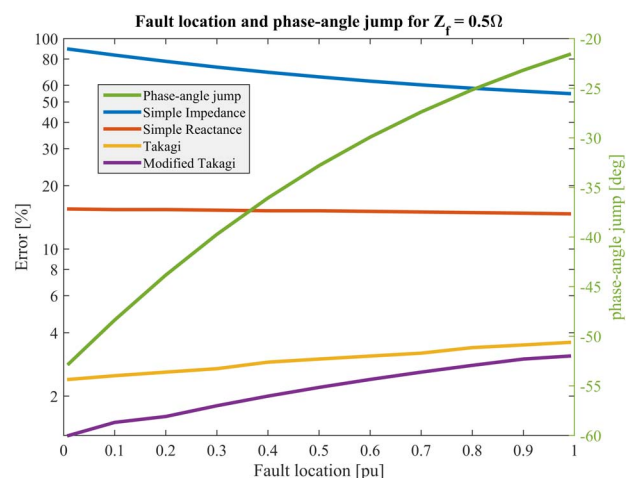


Fig. 3. Fault location and phase-angle jump for fault impedance of 0.5Ω .

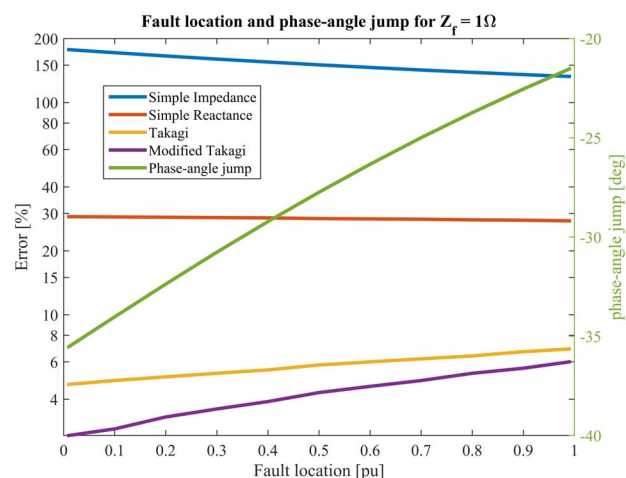
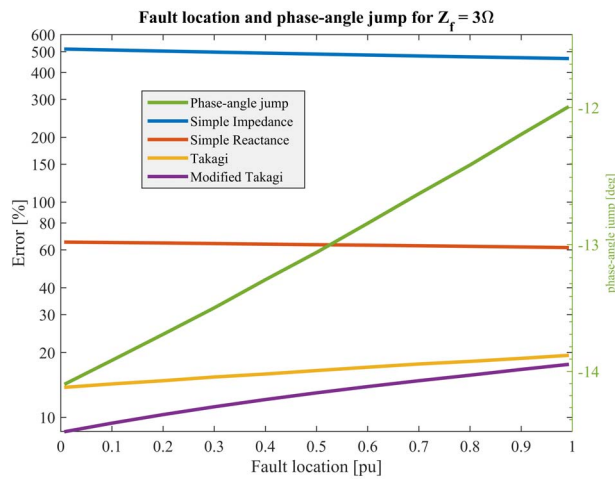


Fig. 4. Fault location and phase-angle jump for fault impedance of 1Ω .

20°) in all cases. In this case, it is evident the influence of phase-angle jump in the increase on the error of all algorithms, since the fault impedance increase is small.

3) *Fault location accuracy with $Z_f = 1\Omega$* : The results of this case are shown in Fig. 4. From Fig. 4, the left axis corresponds to the error of fault location algorithms and the right axis correspond to the computed phase-angle jump. As in the previous case, the error is significant for simple impedance and simple reactance methods. In the same way, Takagi and modified Takagi methods are less affected by the introduction of the fault impedance, but the error increases to more than 2%. The phase-angle jump is more than 20° again, and its influence on the error increase is evident.

4) *Fault location accuracy with $Z_f = 3\Omega$* : The results of this case are shown in Fig. 5. From Fig. 5, the left axis corresponds to the error of fault location algorithms and the right axis corresponds to the computed phase-angle jump. The error is significant for simple impedance and simple reactance methods, as it was mentioned in the previous cases. In this

Fig. 5. Fault location and phase-angle jump for fault impedance of 3Ω TABLE I
SIMPLE IMPEDANCE METHOD ERROR [%] VARYING FAULT IMPEDANCE

Location [%]	$Z_f = 0\Omega$	$Z_f = 0.5\Omega$	$Z_f = 1\Omega$	$Z_f = 3\Omega$
1	0.01	89.2	177.5	514.1
10	0.1	83.5	171.7	508.8
20	0.2	78.0	165.7	503.2
30	0.3	73.2	160.2	497.7
40	0.4	69.2	155.2	492.4
50	0.5	65.8	150.5	487.3
60	0.6	62.8	146.3	482.4
70	0.7	60.3	142.3	477.6
80	0.8	58.1	138.7	473.0
90	0.9	56.3	135.4	468.6
99	1.0	54.8	132.7	464.7

case, the error increased to more than 8% for Takagi and modified Takagi methods. The phase-angle jump, in this case, is quite similar to the no-fault impedance case, but the limits are different. Thus, the difference between the two cases can be made by observing both the residual voltage and phase-angle jump.

A summary of fault location error for all methods is shown in Tables I - IV. From Tables I - IV it can be observed the big influence of the influence of phase-angle jump (fault impedance) in the accuracy of impedance-based fault location algorithms.

B. Fault in low voltage side of transformers

Two more radial systems are used to observe the behavior of phase-angle jump in low-voltage and high-voltage sides of the distribution transformer. The systems are the 4-Node test feeders presented in [10]. The first system is a distribution system fed from a Thévenin equivalent with $V_{th} = 69kV$ and $Z_{th} = 0.1568 + j4.9833$ through a 10-mile transmission line. Two loads are fed from a step-down three-winding transformer through an 1-mile and 150 ft lines respectively.

TABLE II
SIMPLE REACTANCE METHOD ERROR [%] VARYING FAULT IMPEDANCE

Location [%]	$Z_f = 0\Omega$	$Z_f = 0.5\Omega$	$Z_f = 1\Omega$	$Z_f = 3\Omega$
1	0.01	15.5	29.0	65.2
10	0.1	15.4	28.9	64.9
20	0.2	15.4	28.8	64.6
30	0.3	15.3	28.7	64.2
40	0.4	15.2	28.6	63.8
50	0.5	15.2	28.4	63.4
60	0.6	15.1	28.3	63.1
70	0.7	15.0	28.2	62.7
80	0.8	14.9	28.0	62.3
90	0.9	14.8	27.9	61.9
99	1.0	14.7	27.7	61.5

TABLE III
TAKAGI METHOD ERROR [%] VARYING FAULT IMPEDANCE

Location [%]	$Z_f = 0\Omega$	$Z_f = 0.5\Omega$	$Z_f = 1\Omega$	$Z_f = 3\Omega$
1	0.01	2.4	4.7	13.8
10	0.1	2.5	4.9	14.3
20	0.2	2.6	5.1	14.8
30	0.3	2.7	5.3	15.4
40	0.4	2.9	5.5	15.9
50	0.5	3.0	5.8	16.5
60	0.6	3.1	6.0	17.1
70	0.7	3.2	6.2	17.7
80	0.8	3.4	6.4	18.2
90	0.9	3.5	6.7	18.8
99	1.0	3.6	6.9	19.4

The parameters of the system can be consulted in [10]. Five faults are simulated in the high voltage line and three faults are simulated in the low voltage lines, and the phase-angle jump was computed in each case. All faults were simulated with fault impedance equal to zero. The results are shown in Table V.

From Table V it can be observed that the phase-angle jump

TABLE IV
MODIFIED TAKAGI METHOD ERROR [%] VARYING FAULT IMPEDANCE

Location [%]	$Z_f = 0\Omega$	$Z_f = 0.5\Omega$	$Z_f = 1\Omega$	$Z_f = 3\Omega$
1	0.01	1.3	2.7	8.6
10	0.1	1.5	2.9	9.4
20	0.2	1.6	3.3	10.3
30	0.3	1.8	3.6	11.2
40	0.4	2.0	3.9	12.1
50	0.5	2.2	4.3	13.0
60	0.6	2.4	4.6	13.9
70	0.7	2.6	4.9	14.8
80	0.8	2.8	5.3	15.7
90	0.9	3.0	5.6	16.7
99	1.0	3.1	6.0	17.6

TABLE V
PHASE-ANGLE JUMP [$^{\circ}$] VARYING FAULT LOCATION FOR SYSTEM 2

Location [%]	Line 1	Line 2	Line 3
1	-13.45		
25	-7.99	-0.69	0.0
50	-5.62	-0.57	0.0
75	-4.35	-0.49	0.0
99	-3.58		

TABLE VI
PHASE-ANGLE JUMP [$^{\circ}$] VARYING FAULT LOCATION FOR SYSTEM 3

Location [%]	Line 1	Line 2
1	-10.52	
25	-3.47	0.01
50	-2.04	0.00
75	-1.44	0.09
99	-1.12	

for faults in the secondary and tertiary of the transformer is close to zero. So, besides the residual voltage value, the phase-angle jump is a useful tool to identify the fault location on the low-voltage side of a transformer.

The second system is a distribution system fed from a Thévenin equivalent with $V_{th} = 12.47kV$ and $Z_{th} = 0.0598 + j0.7974\Omega$ through a 5-mile distribution line. One load and one motor are fed from a step-down transformer through an 100 ft line. The parameters of the system can be consulted in [10]. Five faults are simulated in the high voltage line and three faults are simulated in the low voltage line, and the phase-angle jump was computed in each case. All faults were simulated with fault impedance equal to zero. The results are shown in Table VI.

From Table V it can be observed that the phase-angle jump for faults in the secondary of the transformer is close to zero. So, besides the residual voltage value, the phase-angle jump is a useful tool to identify the fault location on the low-voltage side of a transformer.

IV. CONCLUSIONS AND FUTURE WORK

The effect of phase-angle jump on impedance-based fault location methods in transmission and distribution systems was presented in this paper. It was demonstrated that the phase-angle jump has a big influence on the accuracy of fault location methods and can be a useful feature to improve fault location algorithms. Simple impedance and simple reactance fault location methods are quite sensitive to fault resistance (which is related to a change in the phase-angle jump). Even if the fault impedance has a low value (e.g., 0.5%) the error of the methods mentioned above reach more than 10% for the total line length. Takagi and modified Takagi fault location methods are less sensitive to fault resistance, but the error increases when the fault resistance (phase-angle jump) increases.

In future work, the phase-angle jump is going to be used to detect faults in non-homogeneous lines and lines with lateral feeders. Also, a fault location that takes into consideration the phase-angle jump and voltage ellipse parameters will be proposed. This algorithm could be capable of detecting the fault and computing the fault resistance. Also, the algorithm is going to be tested using real data.

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